

Titan's strong tidal dissipation precludes a subsurface ocean

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The Cassini mission provided unprecedented insights into Saturn's largest moon, Titan, from its atmosphere to the deep interior¹. The moon's large measured response to the tides exerted by Saturn was interpreted as evidence of the existence of a subsurface ocean^{2,3}. This response, twice the value predicted in pre-Cassini studies, has escaped complete explanation. Here we show that the signature of tidal dissipation in Titan's gravity field is not consistent with the presence of an ocean. Our results arise from the detection of this signature through a reanalysis of the radiometric data acquired by Cassini with improved techniques. We found that substantial energy is being dissipated in the interior (approximately 3–4 TW, corresponding to a tidal quality factor $Q \approx 5$), consistent with recent studies of Titan's rotational state⁴. Because the presence of a liquid layer reduces the tidal dissipation generated below it⁵, these new measurements preclude the existence of a subsurface ocean on Titan and are explained by a model in which dissipation is concentrated in a high-pressure ice layer close to its melting point. This model also reproduces Titan's observed rotational state and static gravity field self-consistently, reconciling all available geophysical measurements. Efficient ice shell convection can prevent widespread melting and ocean formation, but a slushy high-pressure ice layer is consistent with expectations⁶, indicating that it probably hosts liquid water pockets. The forthcoming Dragonfly mission to Titan will provide a further test of whether a subsurface ocean exists.

Cassini orbited Saturn from 2004 to 2017, performing several fly-bys of its moons. One hundred and twenty-four fly-bys were dedicated to the largest moon, Titan, the second largest moon in the Solar system after Ganymede, providing unique insights into this body¹. Ten fly-bys were devoted to measuring Titan's gravity field and led to an estimate of Titan's moment of inertia, revealing a weakly differentiated interior⁷ compared with Ganymede. The interior consists of an approximately 600-km-thick hydrosphere and a low-density rocky core that may point to a large fraction of hydrated silicates⁸ and/or organic matter⁹.

Pre-Cassini studies¹⁰ indicated that these gravity measurements would allow testing for the existence of a subsurface ocean through the first estimation of Titan's complex tidal Love number $k_2 = \text{Re}(k_2) + j\text{Im}(k_2)$, which quantifies Titan's gravitational response to the tides exerted by Saturn¹⁰. Because the tidal deformations and associated perturbation in the gravitational potential are largely enhanced by the presence of a global fluid layer, a large $\text{Re}(k_2)$ might indicate the existence of the ocean¹⁰. The same pre-Cassini study¹⁰ noted that substantial viscoelastic deformations in the interior of an oceanless Titan would have the same effect on $\text{Re}(k_2)$. Crucially, these

viscoelastic deformations produce tidal dissipation through shear friction, thus generating an angular (or temporal) delay between the tidal force and the response, quantified by $\text{Im}(k_2)$ (ref. 11). The detection of this phase delay was thus invoked as a criterion to break this model degeneracy and to conclusively shed light on the existence of the ocean¹⁰. The analysis of Cassini radio tracking data allowed the estimation of $\text{Re}(k_2)$ (ref. 2), yielding a high value (0.616 ± 0.067 (ref. 3)) that was interpreted as evidence of the presence of an ocean and challenged earlier modelling efforts that predicted lower values for $\text{Re}(k_2)$, in the range 0.3–0.4 (refs. 10,12). So far, work has not provided a conclusive interpretation of $\text{Re}(k_2)$ and attributed its large value to several possible interpretations, including a high ocean density ($>1,200 \text{ kg m}^{-3}$)^{13,14}, a strong viscoelastic response of the interior or a dynamic response in the ocean to tides^{3,15}. A recent reanalysis of the same dataset yielded a lower $\text{Re}(k_2)$ (0.375 ± 0.060 (ref. 16)) but was still consistent with a global ocean.

Titan's large obliquity ($0.32^\circ \pm 0.02^\circ$)^{17,18} compared with solid-body predictions was interpreted as possible evidence for the existence of a global fluid layer^{18,19}. However, subsequent analyses showed that the observed obliquity can also be produced by a model with large tidal

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deformations²⁰. The existence of a Schumann-like resonance, which can be triggered by the presence of a subsurface ocean²¹, has been questioned by a recent analysis that showed that the data collected by the Huygens probe during its descent to Titan's surface are best explained by mechanical vibrations of the probe's platform²².

No previous analysis of Cassini radio tracking data was able to confidently detect the phase lag between Saturn's forcing and Titan's response, and the observational tests devised in the pre-Cassini studies¹⁰ involving the combination of $\text{Re}(k_2)$ and $\text{Im}(k_2)$ to conclusively establish the presence of an ocean could not be carried out.

A recent analysis of Titan's rotational state derived the phase lag from the offset of the satellite's orientation from the equilibrium configuration expected for a moon in a damped Cassini state⁴. Titan is expected to be in this configuration of equilibrium in which the spin axis and orbit normal both precess about the normal to the Laplace plane (defined as the average orbital plane) at the same rate. The angular deviation from coprecession depends on the dissipation in the interior and yielded a surprisingly large $\text{Im}(k_2) = 0.120 \pm 0.027$ (ref. 4). Here we use the first direct measurement of $\text{Im}(k_2)$ obtained by observing the time variations of Titan's gravity field to confirm this observation and shed light on Titan's interior state and on the orbital evolution of the Saturnian system.

Titan's gravity field and tidal response

We measured Titan's static gravity field and tidal response by reconstructing Cassini's trajectory through the analysis of radio tracking data acquired during the fly-bys dedicated to gravity science. Our analysis used improved processing techniques, including processing of open-loop radio tracking data and a phase-averaging algorithm that was recently developed and applied to process data from Juno and InSight^{23,24} (Methods). The new techniques allow for the reduction of the data noise and uncertainties in the derived geophysical parameters by about 25–30% (ref. 24) (Extended Data Fig. 1).

We measured Titan's gravity field with lower uncertainties compared with previous studies (Extended Data Table 1). Our solution for the real part of k_2 is $\text{Re}(k_2) = 0.608 \pm 0.048$, consistent with the large value previously reported^{2,3} (Fig. 1). The improved precision also allowed the first measurement of the imaginary part of k_2 . The new measurement $\text{Im}(k_2) = 0.135 \pm 0.035$ is 3–4 times larger than the maximum value expected for a body with an ocean and is consistent with the value recently derived from Titan's rotational state⁴. This direct measurement of $\text{Im}(k_2)$ corresponds to a tidal quality factor $Q = 4.5 \pm 1.1$, indicating strong tidal dissipation in the interior of Titan (the Q of the solid Earth and Mars are about 300 (ref. 25) and 90 (ref. 26), respectively).

The gravity field model inferred from the Doppler data is consistent with a relaxed configuration in hydrostatic equilibrium. The ratio between the measured zonal and sectorial harmonics of degree 2 is $J_2/C_{22} = 3.316 \pm 0.051$, close to the expected 10/3 for a body in hydrostatic equilibrium. Applying the Radau–Darwin approximation, we obtained a moment of inertia (MoI) $C/MR^2 = 0.343 \pm 0.001$ (Supplementary Information), intermediate between the previously reported values of 0.341 ± 0.001 (ref. 3) and 0.348 ± 0.030 (ref. 16).

Titan's interior and absence of an ocean

We explored the implications of our new set of geophysical constraints (mass, MoI , $\text{Re}(k_2)$ and $\text{Im}(k_2)$) by using a Bayesian inversion framework previously applied to Europa²⁷. To carry out the observational test proposed in the pre-Cassini studies¹⁰, we built two separate sets of interior models of Titan with and without a subsurface ocean located between low-pressure (ice I_h) and high-pressure ice polymorphs layer (including ice III, V and VI; see Methods). We included a rocky core below the ice layers but no metallic core, as previous studies discarded this possibility⁷. We jointly inverted all measurements with a Markov chain

Monte Carlo (MCMC) algorithm. The parameter space of the MCMC algorithm includes the mechanical and rheological properties of the layers. The tidal response of the body is modelled with a semi-analytical viscoelastic framework²⁸, allowing tidal dissipation to occur in every layer. The rigidity and viscosity of each layer are varied as free parameters (Extended Data Table 2). The viscoelastic structure of the ice I_h shell is modelled with convection scaling laws²⁹. We modelled the viscoelastic response of each layer with the Andrade rheology^{26,30}. However, the effect of the chosen rheology is not important because the tidal forcing period at Titan (15.9 days) is longer than the nominal relaxation times of ice (see Supplementary Information for further analyses of the effects of rheology).

Our modelling indicates that dissipation in the ice I_h shell alone cannot explain the strong tidal dissipation inferred from the radio tracking data (Extended Data Fig. 2). The rocky core could supply the missing dissipation only if its average viscosity is about 10^{12} – 10^{13} Pa s, several orders of magnitude lower than the viscosity expected for rocks under the GPa pressures in Titan's interior, even if substantially hydrated³¹.

Instead, most of the viscoelastic friction occurs in the high-pressure ice layer above the core. The inferred tidal heating can be generated if the high-pressure layer deforms over the tidal timescale (15.9 days) with an average viscosity of approximately 10^{12} Pa s, which is at the lower end of the range of the melting point viscosity inferred for ice V and VI from laboratory experiments^{32,33}. The low viscosity implies that most of the tidal dissipation occurs in this layer that is close to the melting point throughout its depth. This finding is consistent with expectations from simulations of convective heat transfer in this layer⁶.

Because the presence of an ocean reduces the tidal dissipation generated in the layers below⁵, we found that models with a global subsurface ocean cannot fit both $\text{Re}(k_2)$ and $\text{Im}(k_2)$ (Fig. 2). The largest $\text{Im}(k_2)$ that can be produced by models with an ocean is approximately 0.050, corresponding to $Q \approx 12$, which is inconsistent with our observations and with the independent tidal dissipation inferred from rotational dynamics⁴. However, models without an ocean match both $\text{Re}(k_2)$ and $\text{Im}(k_2)$ within their measurement uncertainty (Fig. 2).

The deep interior structure that we infer from our inversion (Extended Data Fig. 3) is consistent with previous studies. The inferred density of the large (radius of $2,026^{+146}_{-147}$ km, 2σ) rocky core is low ($2,591^{+248}_{-227}$ kg m⁻³, 2σ), indicating a large fraction of hydrated materials⁸ or organic material⁹. The thickness of the outer ice I_h shell is large (170^{+17}_{-17} km, 2σ). Recent experimental work³⁴ discovered an invariant thermodynamic point (the cenotectic point) that represents the lowest possible temperature at which a liquid solution is stable regardless of pressure and composition. The implication for icy moons is that a freezing ocean would completely disappear at an invariant pressure (210 MPa), setting an absolute limit for the thickness of the ice I_h shell that, for Titan, is about 167 km. Our results are therefore consistent with this recent discovery. The corresponding thickness of the high-pressure ice layer (accounting for different high-pressure phases) is 378^{+151}_{-150} km (2σ).

Titan's thermal state and orbital evolution

The measured complex tidal response implies that Titan is dissipating orbital energy at a rate of about 4 TW (an order of magnitude larger than radiogenic heating, estimated at about 300 GW (ref. 8)). Most of the dissipation occurs in the high-pressure ice layer (about 3.5 TW). Convection in the outer shell can extract all of the tidal heating without melting the ice, provided that the viscosity of the basal convective layer is less than about 10^{15} Pa s (Methods). This viscosity range is compatible with the viscosity of the convective layer inferred from the inversion (Extended Data Fig. 3).

The results are also consistent with the suggested presence of methane clathrates above Titan's ice shell, which would explain the persistence of methane in the atmosphere despite the observed loss

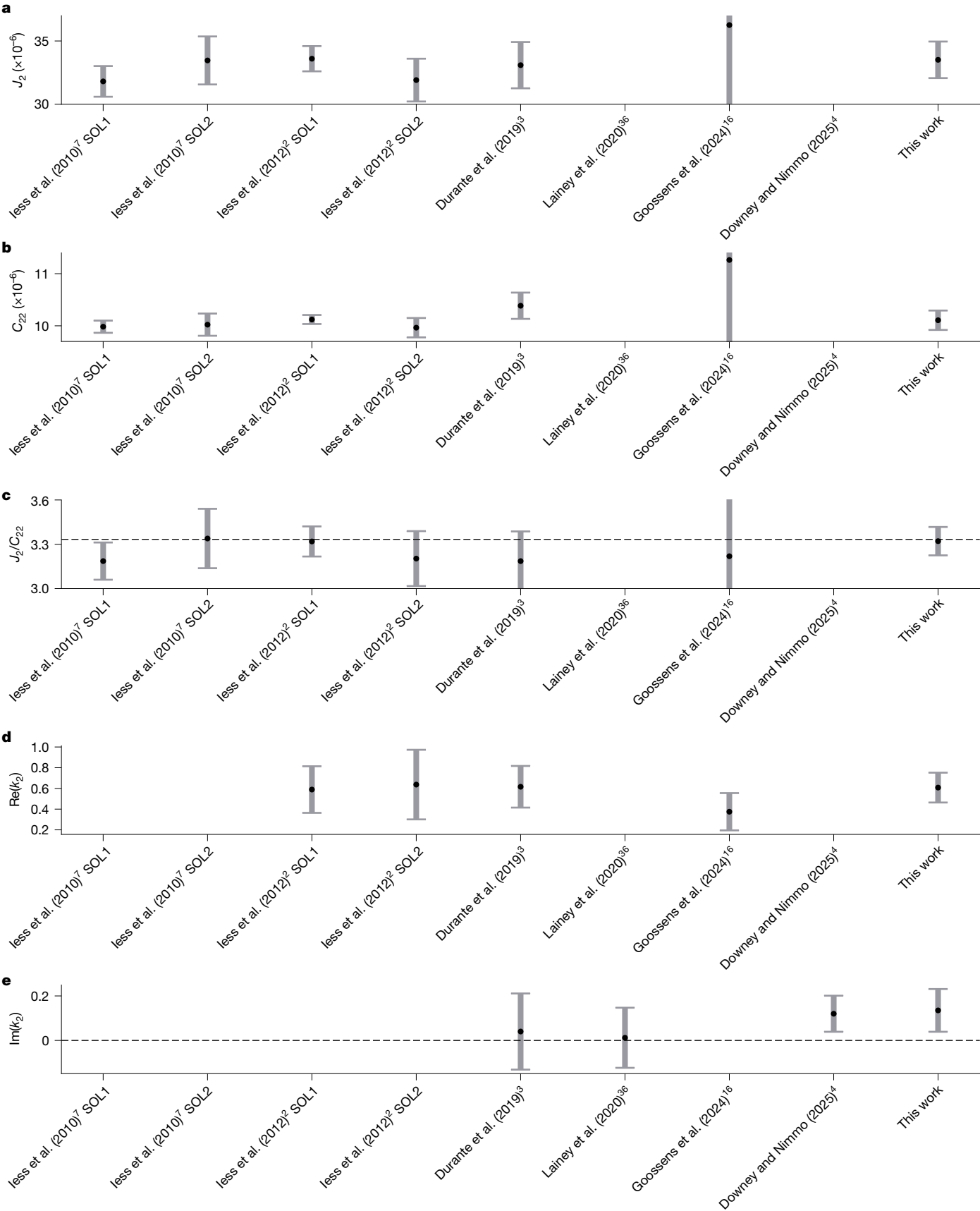


Fig. 1 | Recovered gravity field and tidal Love number k_2 of Titan compared with previous works. The measurements and associated error bars are reported for all of the previous analyses of Cassini radio tracking data and rotational dynamics modelling^{2–4,7,16,37}, including several solutions for some of the studies. The geophysical parameters are J_2 (a), C_{22} (b), J_2/C_{22} (c), $\text{Re}(k_2)$ (d) and $\text{Im}(k_2)$ (e). The dashed black lines represent the value of the ratio J_2/C_{22} (c) expected for a body in hydrostatic equilibrium and $\text{Im}(k_2) = 0$ (e). The error bars represent a 3σ level of uncertainty.

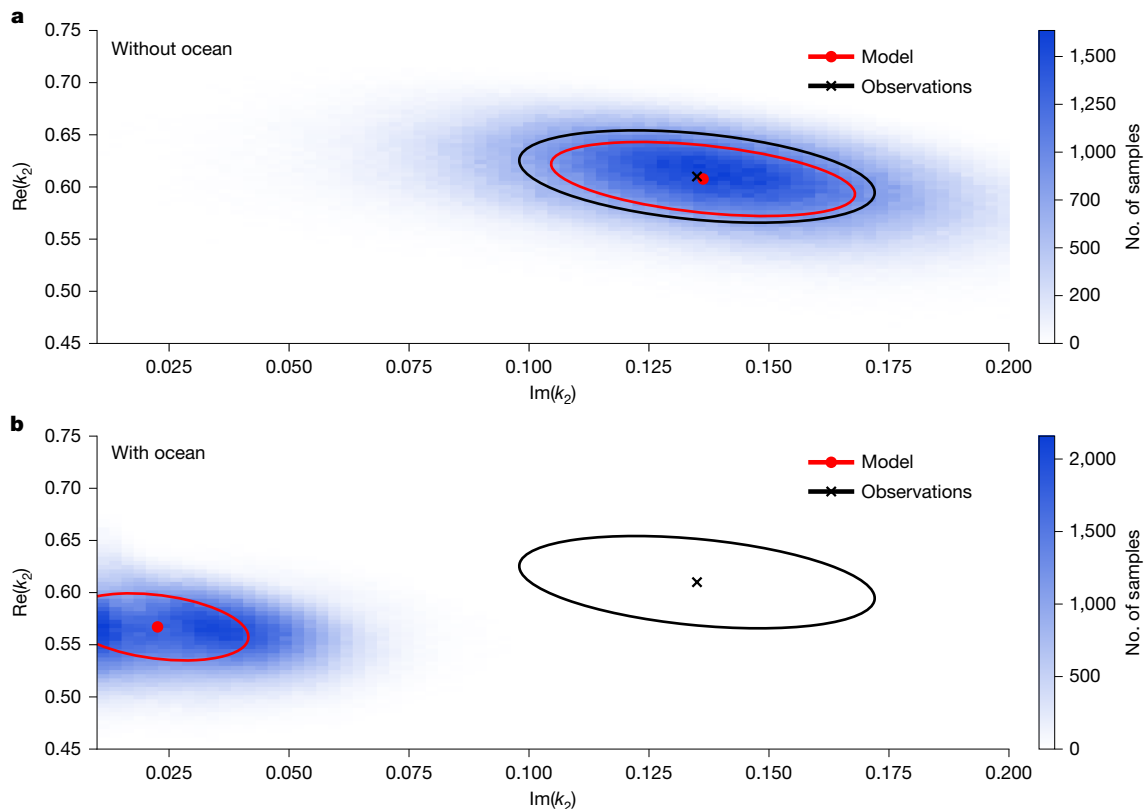


Fig. 2 | Posterior distributions for the tidal Love number k_2 of Titan compared with the observations for models with and without a subsurface ocean.

Histograms and uncertainty ellipse (red) for computed $\text{Re}(k_2)$ and $\text{Im}(k_2)$ compared with the observed uncertainty ellipse (black) for the interior

models of Titan without (a) and with (b) an ocean. The centre and axes of the ellipses are computed from the covariance matrix of the posterior distributions obtained from the interior inversion (1σ level of uncertainty).

to space^{35,36}. Considering the lower thermal conductivity of clathrates compared with ice (about $0.5 \text{ W m}^{-1} \text{ K}^{-1}$), the heat generated by the measured tidal dissipation can be evacuated in the presence of a thermally conductive layer of clathrates 2–3 km thick, which is enough to support a methane atmosphere for several billion years³⁶.

The high value of $\text{Im}(k_2)$ is consistent with the emerging understanding of a rapidly evolving or recently perturbed Saturn system. On the basis of the observed tidal dissipation, Titan's large eccentricity ($e = 0.0288$) should damp to zero on a timescale of approximately 30 Myr (ref. 4). Titan's eccentricity is not primordial, then, and the moon has evolved recently to its current orbital state. Recent astrometric and radio science analyses suggest that Titan formed closer to Saturn and migrated outward owing to strong tidal dissipation in Saturn's interior ($Q_s \approx 125$)³⁷. Titan's current eccentricity, and Saturn's large obliquity and young rings, can be explained by the loss of a satellite in the past approximately 100 Myr, when Titan's outward orbital migration perturbed its orbit³⁸. Alternatively, Titan's eccentricity may have been the consequence of a recent catastrophic collision in the Saturn system³⁹. Considering that the discovery of strong dissipation in the interior of Saturn assumed negligible dissipation in Titan³⁷, our results indicate that Saturn's effective tidal quality factor Q_s at Titan's frequency must be even lower to accommodate for the loss of orbital energy implied by Titan's strong tidal dissipation and match the observed orbital expansion rate ($Q_s \approx 75$; Methods).

Discussion

Our interpretation of Cassini radio science data at Titan is consistent with the existence of a high-pressure ice layer at the bottom of the moon's hydrosphere. So far, high-pressure ices have only been predicted by thermal models^{40,41}. Although this layer was considered a barrier

for energy and material transport across the hydrosphere, our results indicate that intense tidal dissipation occurs in this layer. The inferred low viscosity (about 10^{12} Pa s) would facilitate the onset of strong convection and the subsequent vertical transport of energy and material from the rocks to the surface⁶. It is this convection that is responsible for removing heat rapidly enough that an ocean does not form.

The low viscosity implies that high-pressure ice polymorphs are close to their melting temperature, consistent with expectations⁶. In this state, melt pockets may play a substantial role in reducing the effective viscosity of the ice. Experiments to measure the viscoelastic properties of high-pressure ices subjected to long-term forcing (about 16 days) have historically been challenging to perform, leading to substantial variations in measured rheological responses depending on the technique used^{32,33}. The likely presence of variable grain sizes and impurities further complicates this problem. The primary finding of this study underscores the necessity for new experimental studies using contemporary techniques.

The new results indicate a new pattern in our understanding of large icy moon hydrospheres. The absence of a global ocean in Titan, despite substantial tidal heating, suggests that ocean worlds may be less common than has been supposed in recent years⁴². Although a global ocean has been considered ideal for sustaining habitability in Titan³⁶, the presence instead of a mushy hydrosphere potentially makes this world more interesting. The slushy state of the high-pressure ice layer and the possible widespread occurrence of melt pockets could create sites with highly concentrated organic and saline aqueous solutions, which could be transported upward by strong convection in the ice shell (Fig. 3). These environments could form unique cryoecological niches, enriched by nutrient-rich fluids from convection-driven fluxes originating at the rocky core boundary and surface-derived impact melts transporting complex organic molecules⁴³. Such an environment resembles

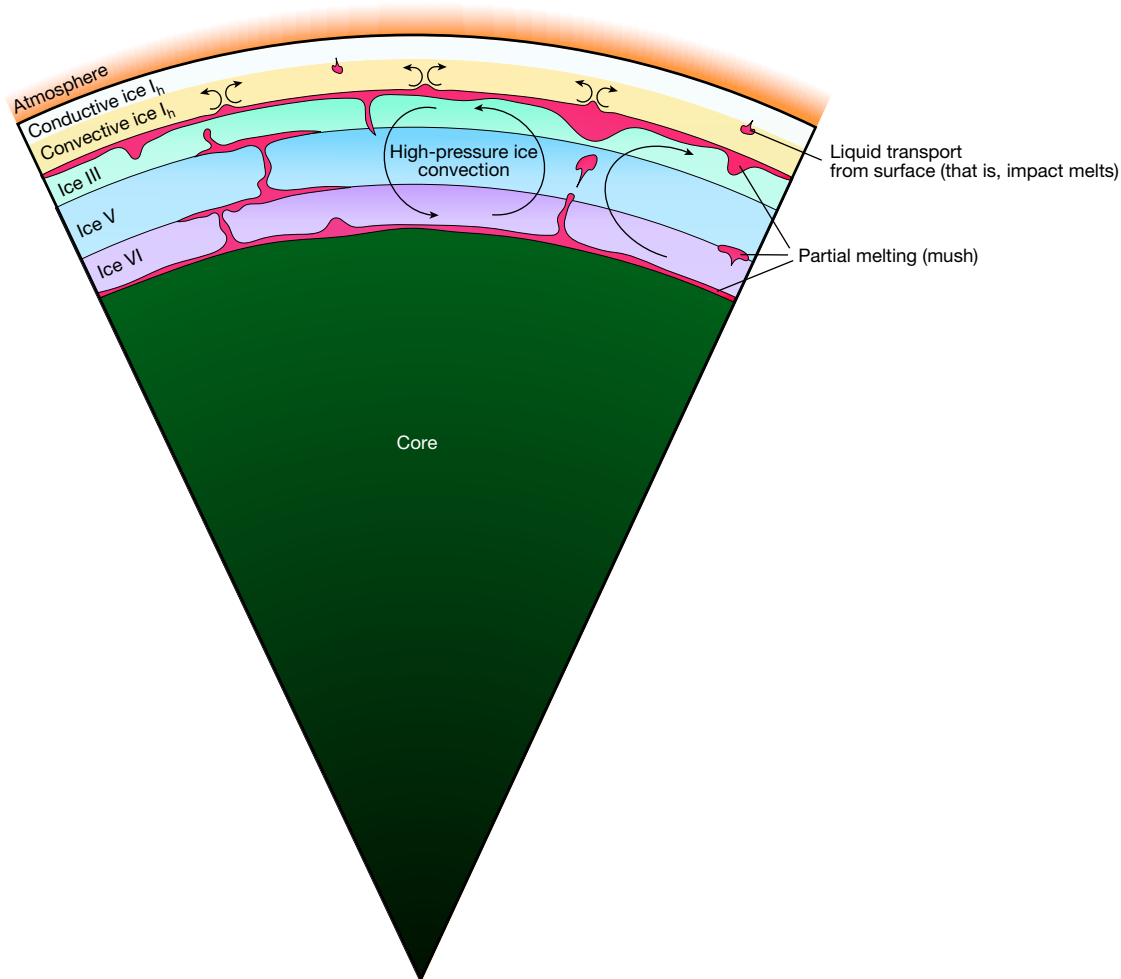


Fig. 3 | Schematic interior structure of Titan revealed by this study. The strong tidal response amplitude and dissipation preclude a global subsurface ocean and indicate a slushy high-pressure ice layer, comprising ice III (light green), ice V (light blue), ice VI (light purple) and small amounts of partial melting (fuchsia).

Earth's polar sea ice ecosystems (one of the largest on Earth), in which a large diversity of organisms thrive despite extreme salinity and near-freezing conditions⁴⁴. Even for conservative melt fractions, Titan's extensive hydrosphere (about $4 \times 10^{10} \text{ km}^3$) would yield enormous liquid volumes—just 1% melt fraction equals the volume of the entire Atlantic Ocean, whereas 0.01% still matches the Mediterranean Sea.

Although both the obliquity and $\text{Re}(k_2)$ have long been interpreted as evidence for a global ocean, reconciling all of Titan's geophysical observations has historically been challenging. Models attempting to explain the obliquity failed to reproduce the observed Cassini plane offset¹⁸ and MoI, requiring MoI values lower than 0.320 (ref. 14), which are largely inconsistent with observations^{2,3,7}. Conversely, models constructed to interpret the Cassini plane offset did not address the obliquity⁴. By contrast, our oceanless interior model is the first model of Titan's interior that simultaneously explains all available geophysical constraints (Extended Data Table 3; see also Supplementary Information). Although measurements of $\text{Re}(k_2)$ remain a powerful tool to detect oceans in small-sized and mid-sized moons, our results demonstrate that measurements of $\text{Im}(k_2)$ are also required to break the degeneracy of possible interior models in the case of large icy moons. An observational test will be provided by the Juice mission, which is expected to measure Ganymede's complex k_2 with excellent measurement precision ($\sigma_{\text{Re}(k_2)} = 10^{-4}$, $\sigma_{\text{Im}(k_2)} = 7 \times 10^{-5}$)⁴⁵.

Because Ganymede—a similar-sized body to Titan—probably has a subsurface ocean⁴⁶, why does Titan not? One possibility is that Ganymede underwent an extreme tidal heating event midway through

its history, melting an ocean that has not yet had time to refreeze⁴⁷. It is likely that Ganymede experienced more tidal heating than Titan, considering that its eccentricity may have been sustained by interactions with the other Galilean satellites in the Laplace-like resonance⁴⁷. The largely distinct differentiation state of the two satellites and the existence of a metallic core inside Ganymede⁴⁶ support this scenario. It is worth considering that a recent analysis of Galileo's and Juno's magnetometer data found a small induction efficiency at Ganymede, interpreted by invoking a thick (roughly 150 km) ice shell⁴⁸. This implies that Ganymede's ocean is thin and close to complete freezing³⁴, despite a comparatively stronger tidal history.

The Dragonfly geophysical investigation including the seismology instrumentation on DraGMet will help constrain the density and rheological structure of Titan, providing an observational test of the new, ocean-free model introduced here with an independent dataset. Revealing the discrete layering of the ice polymorph phases may be possible owing to their substantial seismic velocity contrast⁴⁰, providing a better understanding of Titan's internal heating and the distribution of fluids in its hydrosphere. In the longer term, an orbiter mission⁴⁹ could test for the presence of an ocean by measuring tidally driven elevation changes or rotation rate variations (see Supplementary Information).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information,

acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09818-x>.

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Methods

Radio tracking data selection

We analysed the two-way and three-way Doppler data at 60 s count time acquired by Cassini during the fly-bys devoted to gravity, including T011, T022, T033, T045, T068, T074, T089, T099, T110 and T122, using JPL's Mission Analysis, Operations, and Navigation Toolkit Environment (MONTE⁵⁰). The range-rate observations used to measure the gravity field are extracted from the Doppler shift affecting a radio-frequency signal sent from Earth in X-band (7.2 GHz) and retransmitted back in X-band (8.4 GHz) and Ka-band (32.5 GHz). As in the 2019 analysis³, we selected X/Ka-band data over X/X-band data when available and two-way over three-way data. This approach differs from the data selection used more recently¹⁶, which focused on X/X-band data. However, we also processed X/X-band data for comparison with previous works and found that this choice does not affect the results of the analysis, because the Sun–Cassini–Earth geometry did not imply propagation through the solar corona and the effects of scintillation on different frequencies is negligible. The Doppler data were averaged at 60 s as in the early analyses^{2,3}. We also processed data at 10 s integration time, as in the latest analysis¹⁶. We observed no significant difference in the results. We included the Advanced Water Vapor Radiometer (AWVR) calibrations when available to compensate for the effects of the Earth's troposphere on the radiometric observables. When the AWVR data were not available, we used the Tracking System Analytical Calibration (TSAC) to compensate for tropospheric and ionospheric effects. The data were weighted by the root mean square of the observation residuals individually for each tracking pass.

This work has several improvements in data analysis over previous work^{3,16} involving the application of signal processing techniques developed for Juno⁵¹ and InSight^{23,24}. First, for the closest approach passes, we process the recorded open-loop data from the Radio Science Receiver (RSR) using a phase-locked loop that is optimized for thermal noise⁵¹. This produces unique frequency measurements at 1 s count-time intervals. Second, we apply a phase-averaging technique²⁴ instead of frequency-averaging, as is typically done, to increase the compression time to 60 s. Phase compression is superior over frequency compression for low count times when thermal noise is the dominating source. Phase compression enabled a reduction of data noise of approximately 25% on average, with a peak of 30% for T122, which is consistent with its previous application on InSight radio science data^{23,24} (Extended Data Fig. 1).

Radio tracking data analysis

We modelled the spacecraft dynamics accounting for the gravity accelerations, in a relativistic framework, imparted by the Sun, the planets in the Solar System and all of Saturn's large moons. For Titan and Saturn, we also accounted for the spherical harmonic expansion of their gravity fields. The input gravity field of Titan, expanded to degree and order 5, was initialized to zero and accounted for time-varying tidal effects on degree 2 parameterized by the tidal Love number k_2 . Saturn's gravity field was assumed to be consistent with the ephemerides models⁵². The body-fixed reference frames for Titan and Saturn are also consistent with the ephemerides⁵². We accounted for solar radiation pressure, atmospheric drag using the recently revised Titan Global Reference Atmosphere Model (Titan-GRAM)⁵³ and the acceleration given by radioactive decay in the radioisotope thermoelectric generator with navigation models used by previous studies^{3,16}. We also accounted for the time-varying effects of Titan's atmosphere on its gravity field by expanding the pressure fields obtained from global climate model simulations⁵⁴ into the corresponding surface mass variations^{55,56}. The technique that we used allowed us to take into account both the gravity time variations induced by the direct gravitational attraction of the atmospheric mass and those induced by the response of the moon to the surface loading. The atmospheric mass redistribution induced by

the tides that Saturn exerts on Titan's atmosphere is large, leading to gravity variations on the degree 2 gravity on the order of 10^{-9} , larger than Venus and Mars atmospheric gravity effects by an order of magnitude. By modelling these contributions, we removed their effects from the data and obtained an estimate of k_2 that is only given by the solid moon. These effects are still too small to be detected with fly-bys and they do not markedly affect our solution (Supplementary Information).

The data were segmented into observation arcs centred around the closest approach. We fit the data arcs with a set of parameters describing the dynamics of the spacecraft using a weighted least-squares estimation filter. We integrated the trajectory and computed the Doppler observations. The differences from the raw observations (residuals) were minimized by iterating the filter. A priori uncertainties were used to regularize the least-squares inversion. For each data arc, we estimated the initial position and velocity of Cassini (with a priori uncertainty 100 km and 20 m s⁻¹ for each direction), the atmospheric drag scale factor C_D (a priori equal to 10% of its value) and a scale factor for the solar radiation pressure (10%). As in previous studies^{2,3}, we estimated the accelerations given by the radioisotope thermoelectric generator with a priori uncertainties that are 10^3 – 10^4 times larger than their nominal value. Doppler radio science measurements during fly-by T110 were performed using the low-gain antenna. We estimated the location of its phase centre with respect to Cassini's centre of mass to calibrate the effects of the spacecraft rotation on the Doppler observables. All the information from the arcs is combined to estimate the geophysical parameters, including Titan's gravity harmonics up to degree 5, the tidal Love number k_2 and the associated time lag Δt (from which the imaginary part $\text{Im}(k_2)$ is derived). We also included Titan's gravitational mass when we estimated its orbit. All of these quantities were unconstrained in the inversion, including the higher-degree gravity field that was not constrained with a Kaula's rule.

To be consistent with previous studies, the uncertainties are provided with solutions obtained by numerically integrating and estimating the reference position and velocity of Titan with respect to the Saturn system barycentre at the beginning of each data arc. The a priori uncertainties are given by the covariance of the SAT441 (ref. 52) solution multiplied by 10.

We note that, as acknowledged in the most recent analysis¹⁶, most of the spherical harmonic coefficients at degree 5 are compatible with zero, indicating that the Cassini radio tracking data are not sensitive to full 5×5 expansion. We retained the estimation of the degree 5 coefficients to be consistent with previous analyses, but their inclusion in the data fit is not necessary and it does not yield any further information about Titan's gravity field and interior structure. When we removed these coefficients from the estimation, we were able to further reduce the uncertainties in the geophysical parameters, which supports our finding that both $\text{Re}(k_2)$ and $\text{Im}(k_2)$ are observable through Cassini radio tracking (Supplementary Information).

Uncertainties on estimated geophysical parameters

The reported uncertainties in the Doppler-derived geophysical parameters were obtained from the covariance matrix computed by the weighted least-squares filter. The uncertainties have several sources, the most important being Doppler data noise, the number of observations and the number of estimated parameters. Estimating a larger set of parameters increases the correlations in the solution and the resulting uncertainties. Improvements in the root mean square of the data noise translate almost linearly into improvements in the uncertainties of the retrieved geophysical parameters. When we estimated the Titan ephemerides without using phase compression in our preprocessing, we obtained uncertainties compatible with previous studies: $\sigma_{\text{Re}(k_2)} = 0.060$, $\sigma_{\text{Im}(k_2)} = 0.044$ (refs. 3,16,37). Applying phase compression reduced the data noise root mean square between 20% and 30% for all tracking passes around closest approach, which in turn improved the uncertainties on the derived parameters by about 15–20%:

$\sigma_{\text{Re}(k_2)} = 0.048$, $\sigma_{\text{Im}(k_2)} = 0.035$. The solutions obtained are statistically consistent and the variations are less than 1σ .

We note that the uncertainties in the derived geophysical parameters can be reduced by truncating the estimation of the gravity field up to degree 4 instead of 5, which substantially reduces the number of estimated parameters ($\sigma_{\text{Re}(k_2)} \approx 0.030$, $\sigma_{\text{Im}(k_2)} \approx 0.020$). We thus reported the solution obtained in the most conservative case.

Computing tidal effects on radiometric observables

The least-squares procedure requires computing the partial derivatives of the estimated parameters with respect to the radiometric observables. The recent analysis¹⁶ attributed the notable difference between their estimate of the tidal Love number $\text{Re}(k_2)$ (0.375 ± 0.060) (ref. 16) and previous analyses (0.616 ± 0.067) (refs. 2,3) to an incorrect calculation of the partial derivatives of k_2 with respect to Doppler observations.

The tidal response of Titan is detected through periodic time variations of the gravity field described by variations ΔC_{lm} and ΔS_{lm} in the spherical harmonic coefficients:

$$\Delta C_{\text{lm}} - i\Delta S_{\text{lm}} = \frac{k_{\text{lm}}}{2l+1} \frac{M_{\text{S}}}{M_{\text{T}}} \left(\frac{R_{\text{ref}}}{r} \right)^{(l+1)} P_{\text{lm}}(\sin\theta_{\text{S}}) e^{-im\phi_{\text{S}}} \quad (1)$$

in which k_{lm} are the tidal Love numbers, M_{T} is Titan's mass, M_{S} is Saturn's mass, r , θ_{S} and ϕ_{S} are Saturn's distance, colatitude and colongitude, respectively, with respect to Titan in a body-fixed reference frame centred on the moon, P_{lm} are the associated Legendre functions and R_{ref} is the radius of the sphere to which the spherical harmonic expansion is referred. The tidal lag induced by dissipation is introduced in equation (1) by computing the quantities r , θ_{S} and ϕ_{S} at time $t - \Delta t$, in which Δt is the tidal time lag. The phase lag can be derived as $\delta = \sin(n\Delta t)$, in which n is Titan's mean motion. The tidal quality factor Q is defined as $1/\sin\delta$ and the imaginary part of k_2 is $\text{Im}(k_2) = |k_2|/Q$.

This model is implemented in MONTE through a full variational approach and was used to compute the tidal effects and their partial derivatives. This is the same model that was implemented in the GEODYN code for the latest analysis of Cassini radiometric data¹⁶. We could not find a complete explanation for the differences in $\text{Re}(k_2)$ between our result and the recent analysis¹⁶. Instead, our analysis confirmed previous analyses^{2,3}. To support our results, we provide the results of stability tests that involve different assumptions for the dynamical modelling of the spacecraft and estimation strategy (see Supplementary Information). We found that the estimate of the tidal Love number k_2 is robust under a wide range of assumptions.

Modelling Titan's interior

We modelled the interior of Titan accounting for a rocky core and a hydrosphere that may include an ocean between a high-pressure ice layer and an outer ice I_{h} shell. The radial layers are assumed to be uniform, that is, described by average physical properties. The ice I_{h} shell is further divided into two different layers, the stagnant lid and the well-mixed interior, to account for convection. The number of layers is 4 or 5, depending on whether an ocean is present. The properties of the layers are varied as free parameters in our MCMC algorithm (Extended Data Table 2). The size and density of all layers is explored, except for the radius of the rocky core, which is derived from the body radius and the thicknesses of the other layers. The thicknesses of the ice I_{h} , ocean and high-pressure ices layers are not independent but, instead, are self-consistently constrained by the thermodynamics of the water–ice system given by an equation of state (EoS), which is included in our modelling using SeaFreeze³⁷. To compute the structure of the hydrosphere, we assume a conductive thermal profile in the cold portion of the shell, whereas the warm portion is modelled following scaling laws for convection. For models with a subsurface ocean, the bottom temperature is set by the pressure-dependent ice I_{h} –liquid water phase boundary (melting curve). For models without an ocean, we

select the bottom temperature to require a transition to high-pressure ice. The temperature profile in the remaining high-pressure ice layers is assumed to be the melting curve of each ice polymorph, which are expected to be close to the melting point⁶. This approach has already been used for previous modelling work⁵⁸. This assumption does not affect our results because the other properties of the high-pressure ice layers (density, viscosity, rigidity) are independently varied as free parameters and not tied to any assumption. The EoS implemented in the inversion is based on recent pure water experimental data with higher accuracy and precision than the IAPWS95 formulation in the pressure and temperature range relevant for icy worlds^{57,59}. Accounting for impurities (for example, dissolved ions, clathrate ices) may affect the phase diagram and the resulting hydrosphere structure⁶⁰. We tested this assumption, relaxing the thermodynamic constraint by removing the EoS constraint from the inversion and independently varying the thicknesses of the ice I_{h} , ocean and high-pressure ice layers. Even allowing for thermodynamically forbidden configurations of ocean and ices, models assuming a subsurface ocean still cannot fit the data, whereas oceanless models provide a satisfactory fit of the observations (Supplementary Information).

In the parameter space of the MCMC algorithm, we include the rheological properties of each layer that contribute to the determination of Titan's tidal response, including their rigidity and viscosity. The viscoelastic structure of the ice I_{h} shell is treated by modelling convection with scaling laws²⁹, which include a stagnant lid on top of the warm layer that does not contribute substantially to Titan's tidal dissipation. The stagnant lid thickness is computed on the basis of the other parameters that are varied in the inversion. If this stagnant layer were not included, the whole ice shell would contribute to the viscoelastic response, overestimating the tidal dissipation in that layer. The viscosity of the convective layer is computed using an Arrhenius law based on the modelled temperature structure and the creep activation energy and melting point viscosity varied by the MCMC algorithm. The viscoelastic tidal response is computed with an incompressible semi-analytic framework²⁸, assuming that the layers are described by an Andrade rheological model. The parameters of the Andrade models are also varied in the inversion framework. The parameter α is varied between 0.2 and 0.4 (ref. 61) and ζ is varied between 0.01 and 100, as suggested in recent revisions of the Andrade model^{62,63}. The ocean is modelled as a low-viscosity fluid layer.

The tidal forcing period of Titan (about 16 days) is considerably longer than the nominal relaxation times of ice. Therefore, Titan's tidal response is largely dominated by viscous dissipation, with negligible contribution of the transient anelastic regime (for example, refs. 26,30). As a result, we find that using different viscoelastic models (that is, Maxwell, Burgers and Sundberg–Cooper), which mostly differ in capturing the transient anelastic regime between elastic and viscous regimes near the nominal Maxwell relaxation time, does not affect the results of this study. We provide more detailed assessments of the effects of rheological models and their parameters on Titan's tidal response in the Supplementary Information.

MCMC interior inversion

The joint inversion of all measured geophysical quantities (mass, MoI , $\text{Re}(k_2)$ and $\text{Im}(k_2)$) is carried out by using the MCMC algorithm developed and successfully tested in previous analyses^{27,64}. We used a Metropolis–Hastings sampler to explore the parameter space and build interior models that are consistent with the observations within their measurement uncertainty. The interior properties are randomly sampled and used to evaluate the computed observations $X = [M_{\text{T}}, \text{MoI}, \text{Re}(k_2), \text{Im}(k_2)]$ for each model. The computed values are then compared with the observations X_{obs} with their covariance matrix Σ to build the log-likelihood function $\log(P) = -1/2(X - X_{\text{obs}})^T \Sigma^{-1} (X - X_{\text{obs}})$ that is used to explore the parameter space. The covariance matrix is built with the variances of the observations on the diagonal and it also accounts for the small correlation between $\text{Re}(k_2)$ and $\text{Im}(k_2)$. To validate the

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results obtained with the Metropolis–Hastings sampler, we also ran a separate analysis using the affine-invariant ensemble sampler provided by the open-source library *emcee*⁶⁵, with the same parameter space. The set-up of the algorithm used to validate the modelling approach is based on a previous application to Europa⁶⁶.

Orbital energy and evolution

The energy dissipation rate is computed as¹¹:

$$\dot{E} = -\frac{21}{2} \text{Im}(k_2) \frac{(nR)^5 e^2}{G}, \quad (2)$$

in which n is Titan's mean motion ($n = 2\pi/15.95 \text{ day}^{-1}$), e is the orbital eccentricity, R is the moon's radius ($R = 2,574.7 \text{ km}$) and $\dot{E} \approx 4 \text{ TW}$. The orbital energy loss results in a decrease of the orbital eccentricity at a rate⁶⁷:

$$\frac{de}{dt} = -\frac{21}{2} \frac{k_2}{Q} \frac{M_S}{M_T} \left(\frac{R}{a}\right)^5 ne, \quad (3)$$

in which M_S and M_T are Saturn's and Titan's masses⁶⁸, respectively, and a is the semimajor axis of Titan's orbit. The measured $\text{Im}(k_2) = k_2/Q$ yields a damping rate for the eccentricity of about $3.1 \times 10^{-17} \text{ s}^{-1}$, corresponding to a decay timescale of about 30 Myr.

The semimajor axis of Titan's orbit also decreases at a rate:

$$\left(\frac{da}{dt}\right)_T = -21 \frac{k_2}{Q} \frac{M_S}{M_T} \left(\frac{R}{a}\right)^5 nae^2, \quad (4)$$

Dissipation inside Saturn acts to increase the semimajor axis of Titan's orbit as:

$$\left(\frac{da}{dt}\right)_S = 3 \left(\frac{k_2}{Q}\right)_S \frac{M_T}{M_S} \left(\frac{R_S}{a}\right)^5 na, \quad (5)$$

in which $(k_2/Q)_S = \text{Im}(k_2)_S$ quantifies the dissipation inside Saturn. These two effects compete to determine Titan's orbital evolution. Our measurement of $\text{Im}(k_2)$ has implications for Saturn's tidal dissipation factor Q_S . Recent work measured Titan's orbital expansion rate at approximately 11 cm per year (ref. 37). On the basis of this observation and assuming that Titan's internal dissipation was negligible, the derived Q_S for Saturn was approximately 124. However, Titan's internal dissipation, as described by equation (4), would introduce a negative contribution to its orbital expansion rate, estimated here at approximately -7 cm per year. To reconcile the model with the total observed expansion of 11 cm per year, the positive contribution driven by Saturn's dissipation (governed by equation (5)) must therefore be larger and must account for roughly 18 cm per year. A larger orbital migration rate driven by Saturn implies greater tidal energy dissipation within the planet. Consequently, this revised calculation suggests a lower Q value for Saturn, estimated to be around 75, to match the observed expansion rate of Titan when Titan's internal dissipation is included.

Titan's thermal budget

The heat generation and transfer across Titan's hydrosphere determine the thermal profiles and the physical state of each layer. The implication of Titan's hydrosphere being frozen is that all of the tidally generated heat must be transported through the ice layers and radiated to space without forming sufficient, presumably interconnected⁶⁹, melt in the solid phases to collect and decouple the high-pressure and ice I layers. Because of the strong dependence of ice viscosity on temperature, the thick ice I_h shell (about 170 km) inferred from our model is in the stagnant lid regime. The amount of heat per unit area F that can be removed is ultimately controlled by the stagnant lid developed on top of the convective layer⁷⁰:

$$F = \frac{k\Delta T}{2d} \text{Ra}^{1/3} (\gamma\Delta T)^{-4/3}, \quad (6)$$

in which k is the ice thermal conductivity, ΔT is the temperature drop across the layer (about 170 K), d is the thickness of the layer, Ra is the Rayleigh number that quantifies the strength of convection and γ is a parameter that controls the strength of the temperature dependence of the viscosity of the ice. For bottom-heated shells, $\text{Ra} = \rho g \alpha \Delta T d^3 / (\kappa \eta_b)$, in which $\rho = 920 \text{ kg m}^{-3}$ is the density of the ice, $\alpha = 1.6 \times 10^{-4} \text{ K}^{-1}$ is the thermal expansivity of the ice, $g = 1.35 \text{ m s}^{-2}$ is the gravitational acceleration at the surface, $\kappa = 10^{-6} \text{ m}^2 \text{ s}^{-1}$ is the thermal diffusivity of the ice, $\gamma = E_a / (RT_b^2)$, in which E_a is the activation energy of the ice, $R = 8.3145 \text{ J mol}^{-1} \text{ K}^{-1}$ is the gas constant and T_b is the temperature at the base of the convective layer. Assuming that the thermal conductivity, activation energy and melting point viscosity of the ice vary in the range shown in Extended Data Table 2, up to approximately 10 TW can be removed from the ice shell. This rate of heat removal exceeds that generated by the inferred tidal heating and estimated radiogenic heating, demonstrating that an ocean need not develop. These predictions are all consistent with the constraints of the inversion, which prescribe the viscosity of the ice to be $\lesssim 10^{16} \text{ Pa s}$ to match $\text{Re}(k_2)$ and $\text{Im}(k_2)$. As an example, assuming a bottom viscosity of $5 \times 10^{13} \text{ Pa s}$ yields a convective heat flux of about 4 TW, which is the heat that needs to be evacuated for an ocean not to form. This value also yields $\text{Im}(k_2)$ consistent with observations (Supplementary Information).

Further details are available in the Supplementary Information and rely on refs. 20,71–99.

Data availability

The Cassini radio tracking data used in this research are publicly available through NASA's Planetary Data System at <https://atmos.nmsu.edu/pdsd/archive/data/>.

Code availability

The results of this work can be reproduced using JPL's Mission Analysis, Operations, and Navigation Toolkit Environment (MONTE). MONTE is an astrodynamics programming platform that can be used to perform spacecraft orbit determination. MONTE is ITAR-free and can be licensed for research use. The license for MONTE can be requested at <http://montepy.jpl.nasa.gov>. The code used to compute Titan's tidal response, PyALMA, is available at <https://doi.org/10.5281/zenodo.10472442> (ref. 100).

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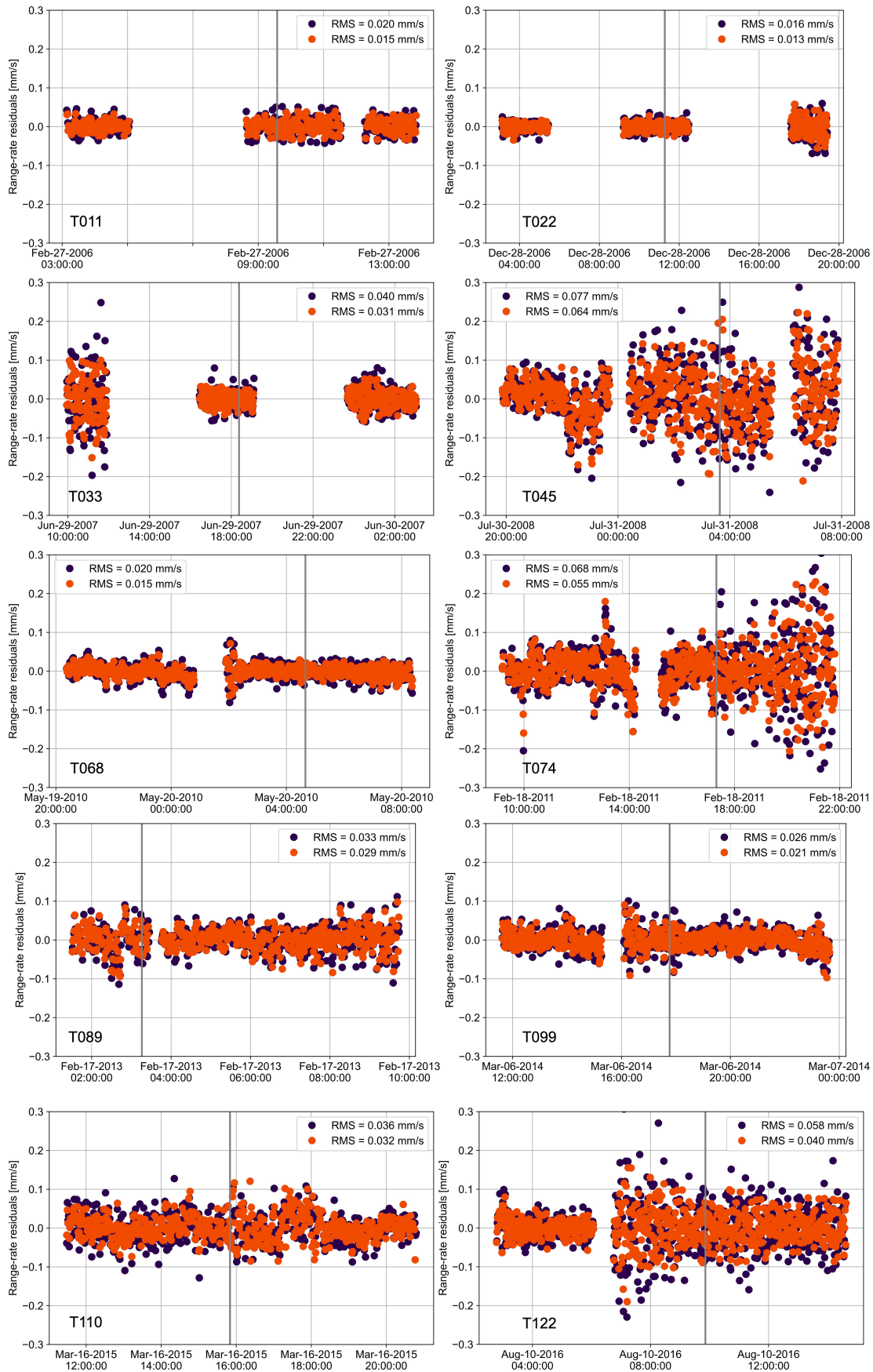
Additional information

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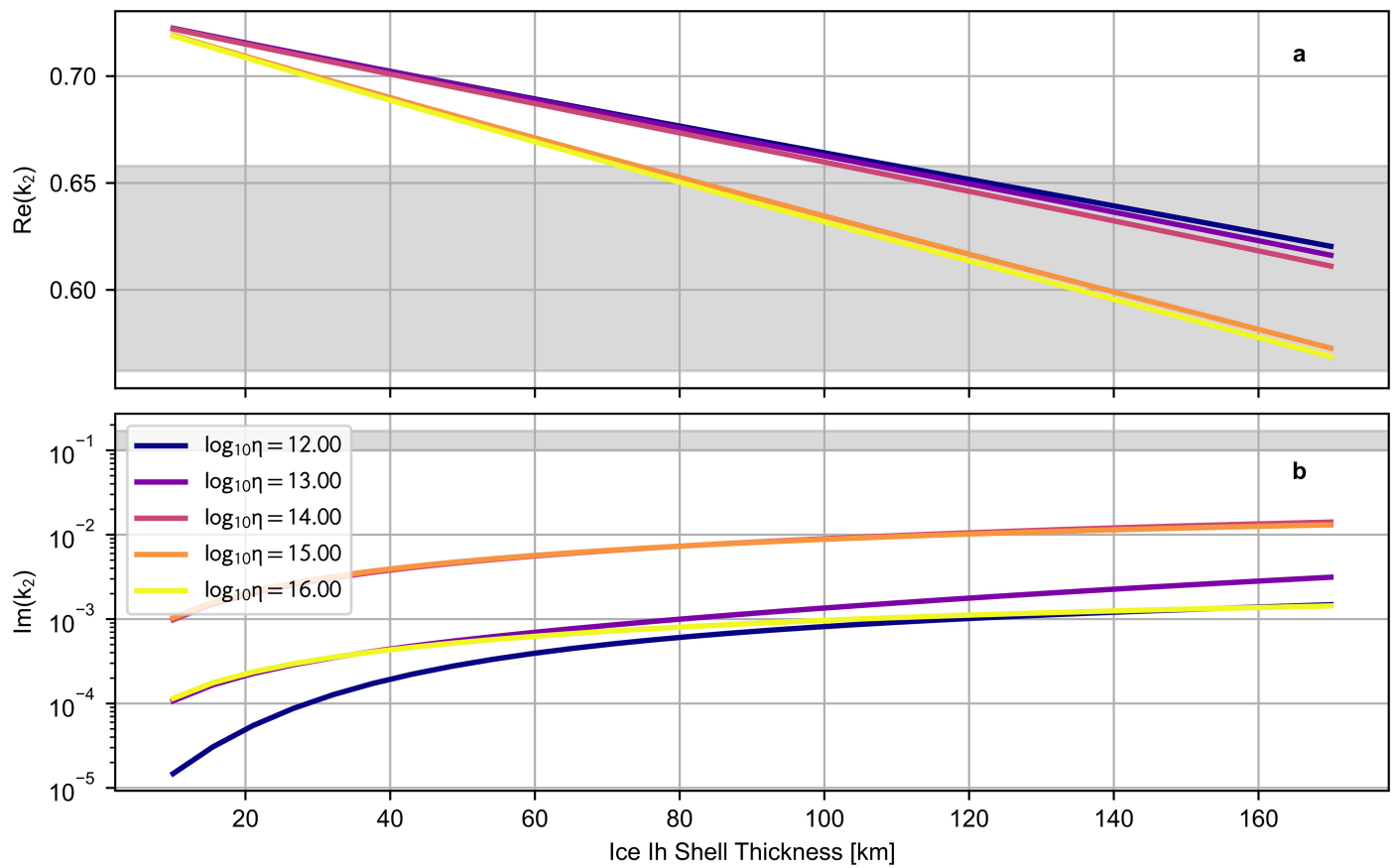
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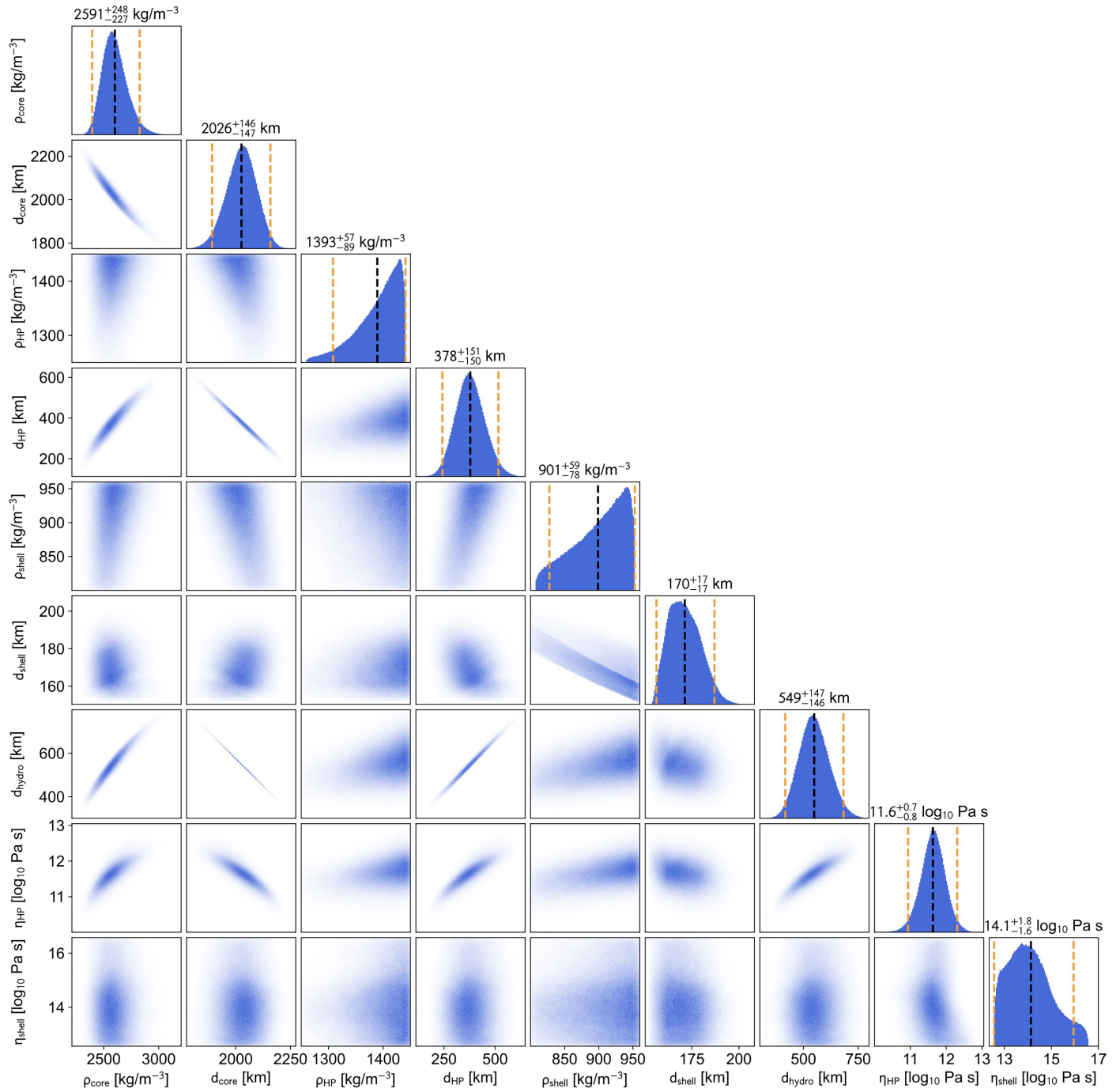


Extended Data Fig. 1 | Range-rate residuals for all of the gravity science fly-bys near Cassini's closest approach to Titan. The orange dots represent the improvement in the data noise allowed by improved processing techniques when compared with traditional techniques (purple dots).



Extended Data Fig. 2 | Complex Love number k_2 for models that assume that tidal dissipation only occurs in the outer ice shell. Comparison between the measured $\text{Re}(k_2)$ and $\text{Im}(k_2)$ and the corresponding modelled values assuming that dissipation only occurs in the ice shell, an ocean density of $1,300 \text{ kg m}^{-3}$

and a hydrosphere thickness of 600 km. The grey-shaded area represents the 1σ measurement uncertainty. The tidal heating generated in the shell alone is not enough to match $\text{Im}(k_2)$, therefore, energy must be dissipated below.



Extended Data Fig. 3 | Posterior distributions of the interior properties resulting from the joint inversion of the geophysical parameters. Corner plot showing the posterior distribution for the interior properties explored with the MCMC algorithm. The properties are the thickness (d), density (ρ) and

viscosity (η) of the layers. Some properties varied in the MCMC algorithm are unconstrained and result in uniform distributions, which are not shown here. The central values are the means of the posterior distributions. The error bars represent a 2σ level of uncertainty.

Extended Data Table 1 | Measured Titan's gravity field

Parameter($\times 10^{-6}$)	This work	Durante et al. (2019)	Goossens et al. (2024)
J_2	33.511 ± 0.471	33.089 ± 0.609	36.256 ± 3.108
C_{21}	0.483 ± 0.116	0.513 ± 0.215	-0.738 ± 0.261
C_{22}	10.107 ± 0.062	10.385 ± 0.084	11.263 ± 1.285
S_{21}	0.695 ± 0.263	0.612 ± 0.359	-1.082 ± 0.625
S_{22}	0.053 ± 0.052	-0.064 ± 0.066	0.515 ± 0.143
J_3	0.992 ± 0.612	-0.179 ± 0.720	0.628 ± 1.199
C_{31}	1.909 ± 0.175	1.481 ± 0.254	1.045 ± 0.346
C_{32}	-0.260 ± 0.129	0.183 ± 0.153	0.720 ± 0.322
C_{33}	-0.233 ± 0.018	-0.222 ± 0.017	-0.113 ± 0.031
S_{31}	1.318 ± 0.323	0.811 ± 0.402	-3.363 ± 0.705
S_{32}	0.391 ± 0.060	-0.027 ± 0.099	-0.337 ± 0.216
S_{33}	-0.231 ± 0.019	-0.226 ± 0.019	-0.099 ± 0.030
J_4	2.056 ± 1.425	-1.077 ± 1.844	-3.074 ± 2.640
C_{41}	-0.473 ± 0.149	-0.842 ± 0.299	-0.285 ± 0.492
C_{42}	-0.007 ± 0.077	0.183 ± 0.107	1.132 ± 0.202
C_{43}	-0.131 ± 0.025	-0.012 ± 0.039	0.031 ± 0.059
C_{44}	-0.011 ± 0.003	0.014 ± 0.003	-0.003 ± 0.005
S_{41}	-0.527 ± 0.546	0.191 ± 0.717	-3.868 ± 1.053
S_{42}	0.513 ± 0.067	0.198 ± 0.106	-0.206 ± 0.157
S_{43}	-0.086 ± 0.030	-0.062 ± 0.033	-0.046 ± 0.051
S_{44}	-0.004 ± 0.003	-0.012 ± 0.004	0.002 ± 0.005
J_5	3.008 ± 1.467	1.118 ± 2.022	-4.762 ± 3.270
C_{51}	0.240 ± 0.268	0.361 ± 0.406	0.340 ± 0.535
C_{52}	-0.056 ± 0.064	-0.097 ± 0.118	0.347 ± 0.184
C_{53}	-0.057 ± 0.012	-0.016 ± 0.019	0.061 ± 0.030
C_{54}	0.004 ± 0.003	0.007 ± 0.004	0.003 ± 0.006
C_{55}	0.000 ± 0.001	0.000 ± 0.001	-0.001 ± 0.001
S_{51}	-0.028 ± 0.481	0.267 ± 0.604	-1.587 ± 0.830
S_{52}	0.173 ± 0.069	0.044 ± 0.094	-0.159 ± 0.156
S_{53}	0.003 ± 0.011	-0.004 ± 0.012	0.067 ± 0.019
S_{54}	-0.005 ± 0.004	-0.002 ± 0.004	-0.019 ± 0.007
S_{55}	0.001 ± 0.001	0.000 ± 0.001	0.000 ± 0.001

Comparison between the gravity field spherical harmonics retrieved in this work and previous analyses (1 σ uncertainties)^{3,16}. The spherical harmonics coefficients are unnormalized and referenced to a sphere with radius 2,575 km.

Extended Data Table 2 | Parameter space of the MCMC inversion

Parameter	Bounds	Unit
Ice Shell Thickness	0-200	km
Hydrosphere Thickness	300-800	km
Ice Shell Average Density	800-960	kg m ⁻³
Ocean Average Density	1050-1250	kg m ⁻³
HP Ice Average Density	1250-1450	kg m ⁻³
Core Density	2000-4000	kg m ⁻³
Ice Ih Shear Modulus	2-5	GPa
HP Ice Shear Modulus	5-9	GPa
Core Shear Modulus	20-50	GPa
Ice Ih Melting Point Viscosity	10 ¹² - 10 ¹⁶	Pa s
HP Ice Viscosity	10 ¹⁰ - 10 ¹⁸	Pa s
Core Viscosity	10 ¹⁸ - 10 ²²	Pa s
Ice Ih Thermal Conductivity	2 - 4	W m ⁻¹ K ⁻¹
Creep Activation Energy	50 - 70	kJ mol ⁻¹
Andrade α	0.2 - 0.4	-
Andrade ζ	10 ⁻² - 10 ²	-

Interior properties and associated bounds considered in the inversion of the geophysical measurements. The values for each property are randomly and freely varied to explore the parameter space and build models consistent with the measurements.

Extended Data Table 3 | Summary of geophysical constraints addressed in relevant interior models of Titan

	MoI	Re(k_2)	Im(k_2)	Obliquity	Cassini Plane Offset
Iess et al. (2010)	✓	—	—	✗	✗
Baland et al. (2011)	✗	—	—	✓	✗
Iess et al. (2012)	✓	✓	—	✗	✗
Baland et al. (2014)	✗	✓	—	✓	✗
Mitri et al. (2014)	✓	✓	—	✗	✗
Durante et al. (2019)	✓	✓	—	✗	✗
Downey and Nimmo (2025)	✓	✗	✓	✗	✓
This study	✓	✓	✓	✓	✓

The tick symbol indicates that the interior model proposed in the corresponding work successfully explained the observation, the cross symbol indicates that the study failed to explain the observation or that the consistency with the observation was not explicitly addressed and the dash symbol indicates that the measurement was yet not available.